

AGNIBH DASGUPTA

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Education

University of Nebraska Omaha, Ph.D., Computing & Information Science	2021 – 2026
Utah State University, M.S., Computer Science	2018 – 2020
West Bengal University of Technology, B.Tech., Computer Science	2013 – 2017

Skills

Research Areas:	Representation Learning, Watermarking, Computer Vision, NLP, Multimodal Learning
Architectures & Models:	Transformers, ViT, CLIP, LLMs, Diffusion Models
Frameworks:	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn
Programming:	Python, Bash, CUDA, Git, Linux, SQL
Distributed:	PyTorch DDP, DeepSpeed, FSDP, LoRA, Slurm/HPC
Post-Training:	SFT, DPO, GRPO, TRL, RLHF
Agentic AI:	LangGraph, RAG, Prompt Engineering
Evaluation:	LLM-as-Judge, RAGAS, A/B Testing, Human-in-the-Loop
Deployment:	vLLM, Quantization, FastAPI, Docker, Railway, GitHub Actions, CI/CD

Awards

CVPR Compute Transparency Champion Award	2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition	
• Awarded to TIACam for exemplary reporting of computational resources, methodology, and reproducibility details	

Professional Experience

University of Nebraska Omaha, Graduate Researcher	2021 – 2026
Representation learning, watermarking, and LLM research	
Utah State University, Graduate Researcher	2018 – 2020
Deep reinforcement learning research for financial trading applications.	

Selected Research and Projects

Path of Exile Q&A Agent	2026
• Wiki-grounded RAG for Path of Exile: live wiki retrieval with multi-query fusion and cross-encoder reranking	
• LangGraph planner orchestrates multi-search as needed before synthesis; FastAPI backend and React trace UI	
• Evaluated with labeled gold pages and LLM-as-judge scores for faithfulness and relevance	
Invariant Features in LLMs: Geometric Characterization and Model Attribution (under review)	2026
• Identified invariant subspaces in 9 open-source LLMs via contrastive sensitivity decomposition	
• Validated semantic and nuisance components causally through next-token prediction interventions	
• Achieved 90%+ zero-shot model attribution accuracy across base, fine-tuned, and distilled variants	
TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation	CVPR 2026
• Built a zero-watermarking framework combining CLIP image-text features with a learned augmentation module	
• Trained a differentiable auto-augmentor to find camera-like distortions, removing hand-designed augmentation	
• Achieved 95%+ watermark recovery on ImageNet and Visual Genome under real-world camera conditions	
Watermarking Language Models through Language Models	IEEE TAI 2025
• Built a model-agnostic watermarking scheme for LLM-generated text without touching model weights	
• Tested on 25 open-source LLMs including fine-tuned and distilled variants using LoRA and DeepSpeed	
• Achieved 88%+ detection accuracy under paraphrasing, fine-tuning, and adversarial prompt attacks	
Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning	CSCI 2023
• Built a ViT-based cross-attention watermarking system with a self-supervised triplet loss objective	
• Trained the encoder to cluster augmented views while separating different images in latent space	
• Reached state-of-the-art comparable robustness under compound noise, confirmed via ablation	
Perspective Transformation Layer	CSCI 2022
• Proposed a lightweight learnable layer for perspective-robust recognition, replacing fixed pipelines	
• Evaluated on SVHN, Imagenette, and MNIST under viewpoint variation with minimal extra compute	
• Outperformed prior state-of-the-art on perspective benchmarks as a drop-in module	